

Revision Strategy after FADO’s First Submission Reviews

September 2026

1 Overall Assessment

Strengths:

- the problem is practically significant;
- the manuscript is generally well organized and clearly written;
- the method consistently improves fairness metrics while maintaining competitive predictive performance;
- the use of 30 random seeds and paired Wilcoxon tests with Holm correction is rigorous;
- the implementation and experimental configuration are reproducible.

The main disagreement is therefore concentrated around **novelty, technical justification, and experimental breadth**, rather than a fundamental problem with the research question.

2 Priority Revision Plan

2.1 A new set of experiments to establish novelty

Compare:

1. ARANYANI;
2. ARANYANI + ADWIN;
3. ARANYANI + ADWIN + learning-rate adaptation;
4. ARANYANI + ADWIN + routing-temperature adaptation;
5. ARANYANI + ADWIN + both adaptations;
6. Full FADO.

This experiment directly answers the strongest novelty criticism from Reviewers 2 and 3.

3 Add a Comprehensive Ablation Study

The absence of ablations is one of the clearest common weaknesses across the reviews, which was expected by us.

A useful ablation should separately evaluate:

- dual-threshold ADWIN;

Priority	Issue	Why it matters
1	Make the novelty of FADO unmistakable	Reviewers 2 and 3 argue that FADO may be an incremental extension of ARANYANI. Kind of critical
2	Add a comprehensive ablation study	Reviewers explicitly ask which individual components are responsible for the gains. This is expected
3	Expand drift experiments	The COMPAS drift is considered artificial, and reviewers request gradual, recurring, and multiple drift scenarios. Also already discussed as a limitation and already have an idea of how to address it.
4	Justify the reaction controller	The learning-rate, temperature, thresholds, and multipliers currently appear heuristic. This is essentially just an ablation of the controller's components.
5	Address EO optimization directly	The paper derives EO regularization but does not use it for training because of computational cost.
6	Reconsider hyperparameter tuning	Offline NSGA-II tuning may appear inconsistent with the fully online motivation. We are not even using NSGA-II as the model is too heavy
7	Add computational overhead analysis	Reviewers want evidence that the method remains practical for streaming systems. Required math that will improve the theoretical foundation.
8	Discuss interpretability	Oblique decision forests create an important limitation for high-stakes applications. Also must do something about it, im just not sure how yet.

Table 1: Recommended revision priorities.

- learning-rate spike;
- routing-temperature adjustment;
- confidence filtering;
- cooldown;
- annealing;
- fairness monitoring, where applicable.

For each variant, report:

- predictive accuracy;
- Demographic Parity (DP);
- Equalized Odds (EO);
- fairness recovery time after drift;
- number/frequency of controller activations;
- computational cost where feasible.

The goal is to answer:

Which component causes the observed improvement?

This converts FADO from a collection of design choices into a scientifically analyzed method.

3.1 Routing-Temperature Adaptation

Reviewer 3 explicitly asks why lowering τ during drift contributes to fairness recovery. This should be supported by:

- an explanation of how temperature affects routing and adaptation;
- an empirical sensitivity analysis;
- ideally, a small theoretical argument about the expected effect on adaptation.

3.2 Sensitivity Analysis

Rather than presenting values such as $\beta_w = 5$, $\beta_c = 10$, or $\tau_{\text{drift}} = 0.1$ as isolated choices, show that the method is not excessively sensitive to them.

For example, vary:

$$\beta \in \{1, 2, 5, 10, 20\}$$

and evaluate several plausible values of τ_{drift} .

The central question is whether the main conclusions remain stable across reasonable parameter choices.

3.3 Stability and Annealing

Explain why the temporary aggressive adaptation is followed by annealing and cooldown. If a formal convergence proof is not feasible, provide empirical evidence that the controller improves recovery without causing persistent instability.

4 Add a proper runtime and efficiency analysis

The paper already has a strong conceptual argument that it avoids storing and replaying historical samples. Quantify this advantage.

Report, where possible:

- training time per instance;
- memory consumption;
- runtime overhead relative to ARANYANI;
- runtime overhead relative to ARANYANI + ADWIN;
- frequency and cost of controller activations;
- additional cost of EO optimization.

This is especially important because practical viability is part of the paper’s streaming-learning motivation.

5 Discuss Interpretability and High-Stakes Applications

Reviewer 1 points out that oblique decision forests are less interpretable than simple axis-aligned decision rules.

The paper does not necessarily need to abandon its high-stakes motivation. Instead, it should acknowledge that:

- oblique splits can reduce interpretability;
- fairness does not by itself make a model suitable for high-stakes deployment;
- transparency, explainability, and legal requirements remain separate concerns.

This limitation should be presented explicitly in the discussion or limitations section.